AIoT Smart Home ESP32-S3

Simulation-based smart-home prototype for comfort, privacy, security and environmental safety

---

Alberto Reyes Castro
MSc Artificial Intelligence Technology portfolio project

Area	Public evidence
Embedded	Two ESP32-S3 Arduino/C++ room controllers
Simulation	Two public Wokwi room simulations
IoT	Optional Blynk monitoring; disabled by default
Safety	Deterministic gas, heat, occupancy and escalation logic
TinyML	Synthetic dataset + historical Edge Impulse evidence
Quality	Validator, scenario tests, pinned ESP32-S3 CI and deterministic report build

Scope: academic/portfolio prototype. Sensors are simulated, thresholds are demonstration values, and no physical-hardware or safety-certification claim is made.

Hardening V2 - 23 October 2026

1. Current public architecture

The repository implements a Bedroom controller focused on comfort, privacy, lighting and Home/Away security, plus a Kitchen/Living Room controller focused on environmental monitoring and staged safety escalation.

Subsystem	Inputs	Outputs / decisions
Bedroom	PIR, indoor/outdoor light, DHT22, buttons	Blinds, ambient/manual lighting, AC, LCD, buzzer, Home/Away
Kitchen/Living	MQ2 score, DHT22, PIR, light, buttons	Gas/heat state, blinds, AC, alerts, emergency-cooling indicator
Optional cloud	Blynk virtual pins	Monitoring and selected controls
Optional ML	gas_score, temperature, humidity	Five-class Edge Impulse prediction when an actual generated runtime is enabl

Both sketches keep `ENABLE_BLYNK = 0` by default, so placeholder cloud credentials cannot block the local Wokwi demo.

2. Bedroom module

- Normal, Sleep, Privacy and Eco modes; Standard, Elderly and Child profiles.
- PIR occupancy, adaptive blinds/lighting, DHT22 thermal control and Home/Away escalation.
- Physical Wokwi outdoor LDR is authoritative by default; an explicitly enabled Blynk V3 can take control until the physical LDR moves again.
- Visible PIN/secret-word values are demonstration-only state-transition controls.

3. Kitchen / Living Room module

The deterministic environmental state is based on project-specific demonstration boundaries plus persistence/warm-up logic.

Synthetic/demo condition	Label
gas_score >= 850 and temperature >= 55 C	GAS_HEAT_EMERGENCY
gas_score >= 850	GAS_DANGER
temperature >= 55 C	HEAT_ANOMALY
gas_score >= 420	COOKING_FUMES
otherwise	NORMAL

These are not calibrated domestic safety limits.

ML-disabled semantics

> **The public Wokwi build keeps `USE_EDGE_IMPULSE_RUNTIME = 0`. In that mode no classifier runs, no model confidence is fabricated, the environmental state is rule-derived, ML confidence is shown as N/A, and raw ML prediction as DISABLED.**

If a reviewer supplies a compatible generated Edge Impulse library and enables the runtime, a valid ML prediction may only increase the selected environmental severity; it does not suppress a more severe deterministic state.

4. TinyML experiment and evidence

The checked-in dataset is simulated and rule-labelled from the same gas/temperature boundaries used by the demo. Humidity is an input feature but does not determine the synthetic ground-truth label. Strong model performance is therefore expected and is workflow evidence, not independent real-world predictive validation.

Historical Edge Impulse run

Item	Recorded evidence
Inputs	gas_score, temperature, humidity
Task	Five-class tabular classification
Training config	100 cycles; learning rate 0.0005; small dense classifier
Training accuracy	~93.4%
Held-out test accuracy	~85.9%
Weighted precision / recall / F1	~0.91 / 0.89 / 0.89
AUC	~0.99 (platform-reported)
Estimated embedded cost	~1 ms; ~1.4 KB RAM; ~14.4 KB flash

The generated inferencing ZIP is intentionally not redistributed. The historical screenshots/model card document the experiment, but the repository does not claim bit-for-bit reproduction of the original hosted Edge Impulse run.

5. Reproducibility and CI

Gate	Purpose
Repository validator	Dataset rules, diagrams, placeholders, token patterns, offline defaults, dependency/document invariants
Scenario contracts	Environmental boundary cases, severity ordering and profile-sensitive gas thresholds
ESP32-S3 compile	Both sketches build against pinned toolchain/library versions
Report consistency	The PDF is rebuilt deterministically and compared with the committed report

Pinned build environment

Component	Pinned version
Arduino CLI	1.5.1
ESP32 Arduino core	3.3.12
Blynk	1.4.0
ESP32Servo	3.2.1
LiquidCrystal_PCF8574	2.3.0
DHT sensor library	1.4.7
ReportLab	5.1.0

The LCD dependency is architecture-agnostic rather than the AVR-declared library that previously generated an ESP32 compatibility warning.

6. Security, privacy and limitations

- Blynk template/auth values are placeholders; Wokwi-GUEST is a simulator network.
- The local room logic does not require Blynk; cloud integration is optional.
- No real household monitoring dataset or private credentials are included.
- The generated Edge Impulse deployment archive is not redistributed; third-party components retain their own licence terms.
- No calibrated gas concentration, certified alarm behaviour or physical ESP32-S3 deployment is claimed.
- Real deployment would require calibrated sensors, independently labelled data, target latency/RAM/flash/power measurements, fault/drift tests, secure provisioning, false-positive/false-negative analysis and domain safety review.

7. Future work

- Validate on physical ESP32-S3 hardware.
- Collect real independently labelled environmental data under a documented protocol.
- Measure deployment resources and failure modes on target hardware.
- Only then reconsider ML-assisted safety decisions beyond the current transparent experimental role.

8. Selected references

Al-Fuqaha, A. et al. (2015). Internet of Things: A survey on enabling technologies, protocols, and applications. *IEEE Communications Surveys & Tutorials*, 17(4), 2347-2376.

Dutta, L. and Bharali, S. (2021). TinyML meets IoT: A comprehensive survey. *Internet of Things*, 16, 100461.

Merenda, M., Porcaro, C. and Iero, D. (2020). Edge machine learning for AI-enabled IoT devices: A review. *Sensors*, 20(9), 2533.

Shi, W. et al. (2016). Edge computing: Vision and challenges. *IEEE Internet of Things Journal*, 3(5), 637-646.

Stojkoska, B.L.R. and Trivodaliev, K.V. (2017). A review of Internet of Things for smart home: Challenges and solutions. *Journal of Cleaner Production*, 140, 1454-1464.

Warden, P. and Situnayake, D. (2019). *TinyML*. O'Reilly Media.

Reviewer interpretation: the current README, source code, reproducibility notes and model card define the public portfolio claim. Historical academic screenshots/metrics are retained as evidence of the earlier experiment, not as runtime telemetry from the hardened Wokwi build.

Repository: github.com/albertoreyescastro/AIoT-smart-home-esp32-wokwi

Wokwi Bedroom: wokwi.com/projects/465661521735358465

Wokwi Kitchen/Living: wokwi.com/projects/465661368262601729